

TokenPowerSandbox: Evidence-Gated CPU-First Screening for Energy-Aware LLM Serving

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Abstract

Energy-aware LLM serving requires comparing configurations under realistic request shapes, yet exhaustive target-GPU profiling is costly and a cheap predictor can be dangerously confident outside its measured scope. We present TOKENPOWERSANDBOX, an evidence-gated workflow that combines an interpretable CPU-resident projector, short target-GPU probes, full-workload verification, and tamper-evident freeze-before-measurement provenance. On one NVIDIA H100 80GB serving Qwen2.5-7B-Instruct with vLLM, three anchor repeats and six development workloads calibrate workload transfer. The same frozen model is evaluated on a blind holdout and a separately predeclared no-refit confirmation totaling 51 post-freeze runs. Energy MAPE is 6.23% and 7.35%, with Spearman rank correlations of 0.976 and 0.933. However, a predeclared TTFT gate passes at concurrency four (9.27% MAPE) and triggers abstention below four (64.80%), showing why energy accuracy cannot certify latency.

A separate 12-candidate study contributes 72 balanced Probe and Verify runs. CPU screening preserves energy rank (Spearman 0.884) but nearly reverses TTFT rank (-0.968) and misses the verified frontier within the frozen 12-candidate set. A retrospective fidelity comparison shows that short measured probes recover that frontier with 1/1 recall (100%) and 1/3 precision (33.3%); full verification identifies the frozen set’s single SLO-feasible Pareto configuration. An independent five-pair confirmation, frozen after selection, finds a mean paired device-energy saving of 1.39% (95% paired-bootstrap percentile interval [1.19%, 1.59%]) and a median paired reduction of 21.43% in P95 TTFT relative to the predeclared expert baseline. Across 154 real H100 serving executions, the central result is not that CPU code simulates a GPU: cheap screening is useful only when evidence labels, scope-aware abstention, and independent verification are part of the system. The present report establishes the single-GPU evidence substrate; adaptive agent policies and multi-GPU transfer are intentionally left to subsequent work.

Keywords

LLM inference, energy efficiency, GPU measurement, multi-fidelity screening, measurement provenance, service-level objectives

1 Introduction

Large language model inference is a continuous operational workload whose energy cost depends on more than parameter count. Input and output token lengths, concurrency, batching, cache policy, precision, and parallelism alter device power, duration, throughput, time to first token (TTFT), and time per output token (TPOT) [7, 14, 16]. A deployment decision must therefore optimize energy under explicit service-level objectives (SLOs), rather than report energy in isolation.

The straightforward method is exhaustive measurement: instantiate every candidate on the target accelerator and run the complete

target workload. This is trustworthy but expensive when accelerators are scarce or queue delayed. Analytical models, simulators, and short profiling runs can screen the space more cheaply [1, 11, 20]. Their usefulness, however, creates a second question: *what evidence is sufficient to act on a prediction?* A model can have low aggregate error while misordering candidates near an SLO boundary, and a workload model calibrated on one utilization regime can fail sharply at another.

TOKENPOWERSANDBOX addresses this trust boundary. It is not a virtual GPU and does not promote a CPU estimate to hardware evidence. It is an execution and evidence workflow with three evidence stages: a deterministic CPU *Sandbox*, a short measured *Probe*, and a complete target-workload *Verify* run. Every retained record carries its acquisition kind, scope, cost, hashes, software and hardware identity, and chronology. Predictions and experiment order are frozen before a blind campaign; unsupported metric-workload regions produce abstention rather than silent extrapolation.

This report consolidates the completed single-GPU evidence into four bounded contributions:

- An evidence contract that separates workload scenarios from tunable serving configurations and prevents simulated, measured, and verified records from being conflated.
- An interpretable CPU-first workload model, calibrated only on six development workloads, followed by a frozen blind H100 holdout and a separately predeclared no-refit confirmation with 51 post-freeze measurements.
- A predeclared scope gate that accepts energy transfer but exposes and withholds inaccurate low-concurrency TTFT predictions.
- A 12-candidate configuration corpus with 72 balanced Probe/Verify runs and a separate 10-run confirmation of the selected configuration.

The evidence spans 154 real H100 serving executions including three calibration anchors. It supports same-H100 workload screening and a fixed single-GPU configuration case study. It does not establish cross-model, cross-GPU, multi-GPU, or multi-node transfer, nor does it evaluate an adaptive LLM agent. Those boundaries are deliberate: the paper isolates the evidence substrate on which a later TokenPowerAgent can safely operate.

2 Problem, Evidence, and Claim Boundary

An inference scenario is

$$S = \langle M, W, H, R, C, \mathbf{o} \rangle, \quad (1)$$

where M identifies model and precision, W fixes request and token geometry, H identifies the base target hardware and fixed operating envelope, R identifies the serving runtime, C contains memory and SLO constraints, and $\mathbf{o}$ lists the objectives. A serving configuration $x \in X_S$ contains only deployable controls such as maximum sequences, batched-token budget, cache and prefill

policy, tensor/pipeline/data parallelism, and power cap. Keeping S and x separate prevents an apparent optimization from silently reducing the requested work.

For evidence stage $e \in \{\text{SANDBOX}, \text{PROBE}, \text{VERIFY}\}$, let $f^{(e)}(S, x)$ contain GPU device energy per 1,000 output tokens and output-token throughput. Verify executes the full target workload and is the reference fidelity. Under minimization, the frozen configuration study uses

$$f^{(\text{VERIFY})}(x) = \langle E_{1k}, -\Theta \rangle. \quad (2)$$

P95 TTFT and P95 TPOT determine feasibility and are also reported, but they are not additional Pareto objectives in the frozen scenario. Candidate x is feasible only if its verified metrics satisfy every SLO. The target set is the nondominated subset of the frozen verified feasible candidates.

Every backend emits a typed record for the campaign log,

$$\mathcal{R} = (h(S), h(x), e, \kappa, y, u, c, z), \quad (3)$$

where $h(\cdot)$ is a SHA-256 identity. The field κ records one of three acquisition semantics: simulated, measured, or verified. The vector y holds outcomes, u is declared uncertainty, c is realized GPU-hour cost, and z stores execution and provenance. A topology-aware CPU calculation remains simulated or extrapolated regardless of sophistication. Only real hardware execution can create measured or verified evidence.

Figure 1 summarizes the contract. The current claim is narrower than “predict any deployment.” We test whether cheap evidence can screen workloads and configurations within one measured H100 envelope, whether its failure is detectable, and whether a candidate selected through that ladder survives an independent target-workload confirmation.

3 TokenPowerSandbox Design

3.1 CPU-Resident Projector

The base projector rejects memory-infeasible configurations, separates prefill and decode work, scales a measured anchor by token geometry and batch occupancy, and estimates active power between measured idle and active bounds. It preserves the physical identity

$$\widehat{E}(S, x) = \widehat{P}(S, x) \widehat{t}(S, x), \quad (4)$$

then derives energy per 1,000 output tokens, throughput, TTFT, and TPOT with a term-by-term decomposition.

One anchor cannot identify utilization transitions. Relative to anchor workload (L_0, k_0) , define

$$q = \log_2 \frac{L_{in} + 0.5L_{out}}{L_{in,0} + 0.5L_{out,0}}, \quad r = \log_2 \frac{k}{k_0}, \quad (5)$$

$$\phi = [q, r, qr, r^2]. \quad (6)$$

For directly predicted metric m , ridge regression fits

$$\log \frac{y_m}{\widehat{y}_m^{base}} = \beta_m^T \phi. \quad (7)$$

The ridge strength is selected per metric by leave-one-workload-out MAPE on the development split. We correct throughput, average power, TTFT, and TPOT; energy is recomputed from corrected power and duration rather than independently fitted. The four features and their scales and coefficients are retained in the frozen model profile.

3.2 Measured Probe and Verify Paths

The measured executor launches a digest-pinned vLLM server and a CPU-only benchmark client in restricted containers. Prefix caching is disabled; temperature, seed, output length, and EOS behavior are fixed. A READY/GO/DONE/ACK handshake places reads of DCGM’s cumulative device-energy counter around the active request window, excluding model loading, health checks, graph capture, warmup, and result formatting. The executor emits 100-ms telemetry, benchmark output, server metadata, model and tokenizer revisions, image digest, configuration, and power-limit readback, then seals the emitted-file hashes in a campaign manifest.

Probe and Verify use the same executor and metrics but different workload durations. In the configuration study, Probe submits 64 requests and Verify submits the complete predeclared 256-request workload. Repeats are summarized by median and median absolute deviation (MAD); stage labels and manifest hashes remain attached to the summaries.

3.3 Freeze, Validation, and Scope Gate

Before each post-development campaign, the system serializes predictions, intervals, scenario and configuration identities, workload order, repeat schedule, software hashes, and a file manifest. The validator rejects a comparison if predictions do not precede measurements or if any identity, split label, repeat, power state, or raw-artifact hash differs. Holdout and confirmation data are never accepted as fit input.

The residual model is eligible only for the measured model, runtime, configuration, GPU count, output length, request regime, and bounded context-concurrency envelope. A second release gate may narrow individual metrics inside that envelope. Our final workload campaign predeclares an occupancy-stratified TTFT rule. The decision is generated and hashed without refitting after measurement. Uncalibrated prediction intervals remain labeled diagnostic and cannot be interpreted as statistical coverage.

4 Experimental Methodology

4.1 Platform and Measurement Boundary

All serving experiments run on one NVIDIA H100 80GB HBM3 GPU at a fixed 700-W limit, using driver 580.173.02 and DCGM 4.6.1. A digest-pinned vLLM 0.23.0 container serves Qwen2.5-7B-Instruct revision a09a35458c702b33eeacc393d103063234e8bc28 in BF16. Tensor, pipeline, and data parallel degrees remain one. The study measures H100 device energy during the active benchmark window using DCGM’s cumulative device-energy counter (field 156); host, network, cooling, model loading, and embodied energy are outside the boundary. TTFT and TPOT denote per-run P95 values, summarized across repeats by their median unless a paired effect is explicitly named.

4.2 Workload-Transfer Protocol

The transfer study fixes maximum sequences at 256, maximum batched tokens at 8,192, chunked prefill on, prefix caching off, 128 output tokens, and unbounded offered rate. Request count is eight times concurrency. Three repeats at 512 input tokens and concurrency eight form the measured anchor. Six development workloads

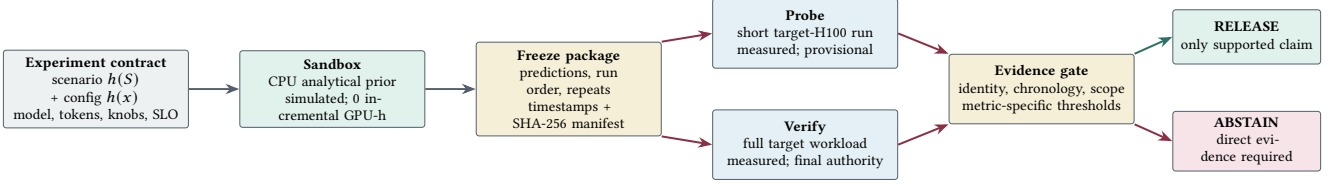


Figure 1: Evidence contract and acquisition paths. Sandbox records are simulated. Predictions, identities, and the balanced schedule are hash-frozen before the interleaved Probe/Verify measurements; the configuration corpus compares fidelities retrospectively. The gate releases only identity-, chronology-, and metric-scope-consistent claims or abstains.

spanning 256–2,048 input tokens and concurrency 1–32, each repeated three times, fit the residual model; leave-one-workload-out energy MAPE is 10.11%.

Holdout v2 freezes eight disjoint workloads before 24 cyclically balanced measurements. Scope confirmation v3 then reuses the unchanged profile on a separately predeclared 3×3 grid, $L_{in} \in \{384, 768, 1536\}$ and $k \in \{1, 2, 4\}$, for 27 additional measurements. Its primary endpoint is all-workload energy MAPE $\leq 15\%$. The v3 boundary test releases TTFT at $k = 4$ only if stratum MAPE is $\leq 20\%$; the rule requires abstention below four when sparse TTFT MAPE exceeds 20%. Support above the tested boundary is reported only together with the previously frozen holdout-v2 evidence at $k \geq 4$.

4.3 Configuration-Fidelity Protocol

The configuration study fixes a 2,048-input/128-output workload at concurrency 32, unbounded rate, and the same model, GPU, precision, and 700-W limit. The SLO requires P95 TTFT $\leq 1,600$ ms and P95 TPOT ≤ 20 ms. Twelve candidates vary maximum sequences (8, 16, 32), maximum batched tokens (2,048, 4,096, 8,192), and chunked prefill; two no-chunk controls and the predeclared expert baseline (256 sequences, 8,192 batched tokens, chunking on) complete the predeclared set.

All CPU predictions and the cyclically balanced acquisition schedule are frozen first. Every candidate then receives three Probe repeats and three Verify repeats, yielding 36 and 36 real runs. The schedule interleaves the two fidelities without using one to select which candidate receives the other. Median Verify records define the SLO-feasible reference set and Pareto frontier within these 12 frozen candidates. We compare stages using MAPE, Spearman correlation, pairwise ordering, frontier precision/recall, and realized GPU-hours.

After the corpus identifies seq32-bt2048-chunk and the expert baseline, we lock both candidates and run a new five-seed, alternating Verify campaign. No refitting or reselection occurs. Its predeclared gate requires all five energy pair wins, a positive lower bound for the mean paired energy saving in a 95% bootstrap percentile interval, lower median paired P95 TTFT, and successful SLO-feasible execution. The interval exhaustively enumerates all 5^5 paired bootstrap resamples.

Table 1: Frozen workload-transfer results. Base values are the uncorrected analytical projector; corrected values use development-only residuals. Pair counts exclude observed ties.

Metric	Holdout v2 MAPE		Corrected rank ρ	Pairs
	Base	Corrected		
Energy/1k output	63.01%	6.23%	0.976	27/28
Output throughput	23.93%	10.53%	0.976	27/28
P95 TTFT	73.44%	32.67%	0.952	26/28
P95 TPOT	34.29%	7.99%	0.929	25/28
No-refit scope confirmation v3				
Energy/1k output	7.35% MAPE		0.933	33/36
Output throughput	11.31% MAPE		0.867	30/36
P95 TTFT	46.29% MAPE		0.800	30/36
P95 TPOT	9.03% MAPE		0.736	28/35

5 Results

5.1 Blind Workload Transfer Preserves Energy Decisions

Table 1 reports the frozen workload studies. On holdout v2, the development-only correction lowers energy MAPE from 63.01% to 6.23%. Median and maximum energy APE are 4.96% and 13.84%; rank correlation is $\rho = 0.976$, and 27/28 pairwise workload orderings are correct. Throughput and TPOT improve to 10.53% and 7.99% MAPE. Median energy repeat CV is only 0.78%, so prediction error, rather than repeat noise, dominates.

Without refitting, scope confirmation v3 obtains 7.35% energy MAPE, 10.93% maximum APE, $\rho = 0.933$, and 33/36 correct energy orderings. It passes the frozen 15% primary endpoint and identifies the same minimum-energy workload as the measurement. Figure 2 shows that all 17 individual energy errors are below 15%; the second campaign separately passes its predeclared aggregate-MAPE gate.

5.2 The Metric-Specific Gate Detects Latency Failure

Energy accuracy does not imply that every component of the prediction is accurate. Aggregate TTFT MAPE is 32.67% on holdout v2 and 46.29% on scope confirmation. The predeclared strata locate the failure. At concurrency four, TTFT MAPE is 9.27% and no point exceeds 14.55%, passing the 20% release threshold. At concurrency one and two, TTFT MAPE is 79.27% and 50.34%; their combined sparse stratum reaches 64.80%. The six holdout-v2 workloads at $k \geq 4$ have 10.66% TTFT MAPE and 14.16% maximum APE. Together, the

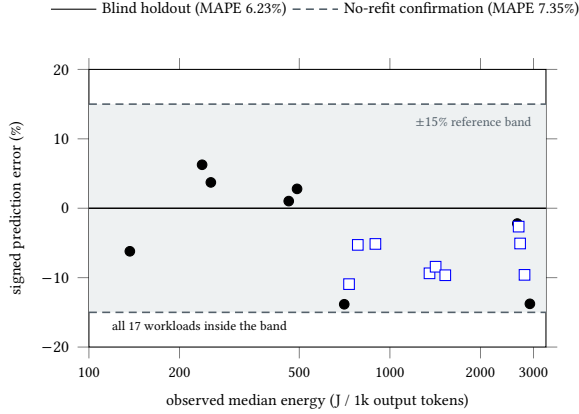


Figure 2: Signed error of frozen CPU energy predictions on the blind holdout and no-refit scope confirmation. Positive values denote overprediction. All 17 workloads remain inside a $\pm 15\%$ band; for the no-refit confirmation, 15% was the predeclared primary endpoint. Rank correlations are 0.976 and 0.933.

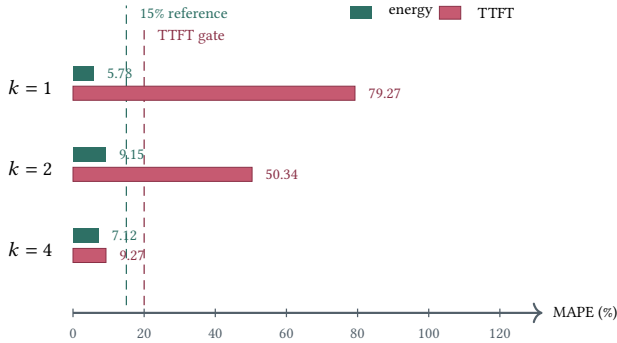


Figure 3: No-refit scope confirmation by concurrency k . The campaign passes its predeclared aggregate energy-MAPE endpoint; observed energy MAPE is also below 15% in each stratum. TTFT passes its 20% boundary test at $k = 4$ but fails below it, triggering explicit abstention.

frozen evidence releases TTFT at $k \geq 4$ inside the present envelope and abstains below four.

Figure 3 makes the asymmetry visible. Energy remains below 10% MAPE in every stratum, yet at $k = 1$ throughput and TPOT errors rise to 24.02% and 18.25%. Compensating power and duration errors can preserve energy while the latency decomposition is wrong. This is precisely the false-confidence case that a single aggregate metric would hide.

5.3 Configuration Screening Requires Measured Probes

All 72 configuration runs succeed and predictions precede measurements. The collection-time validation report records 190 hashed artifact entries as verified. The original per-sample configuration

Table 2: Discovery-corpus median Verify outcomes for the predeclared expert baseline and the unique SLO-feasible Pareto configuration within the frozen 12-candidate set.

Metric	Expert	Pareto	Change
Energy (J/1k out.)	371.13	367.27	-1.04%
Output tok/s	1732.60	1785.63	+3.06%
P95 TTFT (ms)	1343.60	1053.56	-21.59%
P95 TPOT (ms)	17.25	16.76	-2.84%

telemetry archive is unavailable, so those hashes cannot now be rechecked; the 84 retained stage-level replay records permit re-computation of the reported configuration metrics. The CPU configuration records are deliberately conservative negative-baseline extrapolations: the workload residual is not reused after serving controls change, and every record retains `validation_required`. They test whether the analytical prior alone is adequate for configuration decisions.

It is not. The Sandbox preserves useful energy ordering ($\rho = 0.884$) but has 19.45% energy MAPE and nearly reverses TTFT rank ($\rho = -0.968$). If used alone, its stage-local frontier would select the predeclared expert baseline instead of the frozen set’s verified frontier. Its frontier recall is therefore 0% (zero of the one verified Pareto point in this set). A good energy rank does not repair incorrect SLO feasibility.

In the retrospective fidelity comparison, the 64-request Probe sharply improves decision fidelity. Energy and TTFT MAPE fall to 1.15% and 17.20%. Its provisional frontier has three candidates and contains the frozen set’s single verified Pareto point: recall is one of one (100%) and precision is one of three (33.3%). Probe avoids a false negative; full Verify remains necessary to remove two false positives. This corpus does not evaluate an adaptive Probe-then-Verify selection policy. Figure 4 shows both the TTFT rank inversion and the verified objective space.

Among five Verify-feasible candidates, `seq32-bt2048-chunk` is the unique Pareto point within the frozen 12-candidate set under the declared objectives and SLOs. In the discovery corpus, it uses 1.04% less energy, provides 3.06% more output-token throughput, lowers P95 TTFT by 21.59%, and lowers TPOT by 2.84% relative to the expert baseline (Table 2). Its average power is slightly higher, so the device-energy benefit comes from completing the fixed work sooner rather than suppressing instantaneous power.

5.4 Independent Confirmation Survives Reselection Control

The selected configuration is then tested in a new campaign, not re-reported from its discovery repeats. All 10 runs succeed and satisfy both SLOs. The locked candidate wins all 5 seed-matched energy pairs, with mean paired device-energy saving 1.39% (95% paired-bootstrap percentile interval [1.19%, 1.59%], exhaustive enumeration of all 5^5 resamples; one-sided exact sign test $p = 0.03125$). Mean paired throughput increases 3.46%, median paired P95-TTFT falls 21.43%, and mean paired P95-TPOT falls 3.30%. Table 3 reports the exact medians and paired effects.

This result is intentionally local: one model, one request shape, one H100, and two fixed configurations. It demonstrates that the

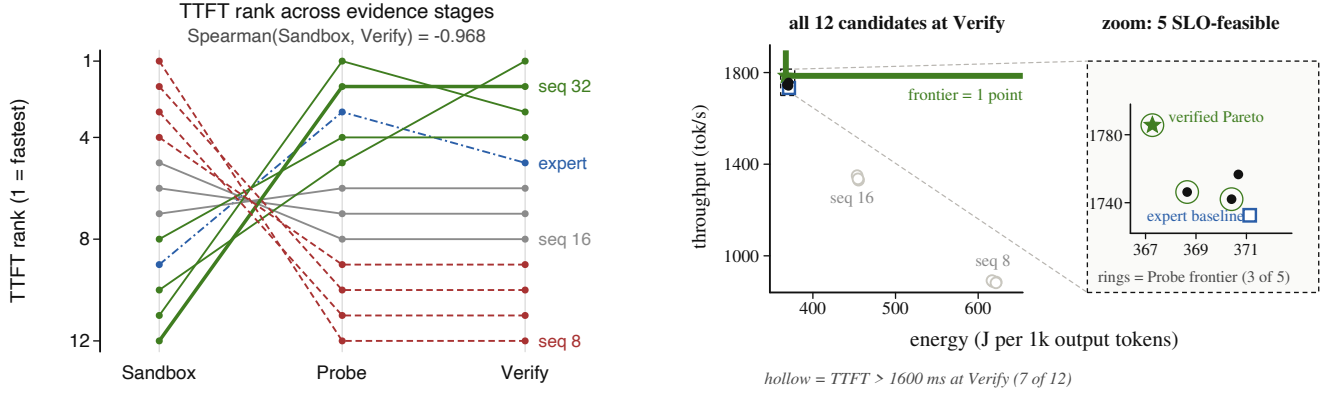


Figure 4: Why full verification remains necessary. Left: Sandbox P95-TTFT rank is nearly reversed relative to Verify ($\rho = -0.968$); a short measured Probe repairs most ordering. Right: Verify medians for all 12 frozen candidates, with an inset magnifying the five SLO-feasible candidates; hollow points are the seven candidates whose TTFT exceeds the 1,600-ms SLO. Rings denote the three-candidate Probe frontier (one of three lies on the verified Pareto set), the square marks the expert baseline on the Sandbox stage-local frontier—the choice if Sandbox were used alone—and the star marks the unique verified Pareto configuration within the frozen 12-candidate set.

Table 3: Independent five-pair Verify confirmation after candidate lock.

Metric	Expert median	Winner median	Paired effect
Energy (J/1k out.)	372.21	367.26	-1.39% mean
Output tok/s	1727.44	1783.71	+3.46% mean
P95 TTFT (ms)	1340.92	1057.11	-21.43% median
P95 TPOT (ms)	17.29	16.77	-3.30% mean

Column medians are descriptive; effects aggregate seed-matched percentage changes. Energy: 5/5 pair wins; mean paired saving 1.39%; 95% paired-bootstrap percentile interval [1.19%, 1.59%] from all 5⁵ resamples; one-sided exact sign test $p = 0.03125$.

candidate identified from the fully verified corpus survives an independent same-scope rerun; it is not a claim of fleet-wide or cross-hardware energy savings.

5.5 Measurement Cost and Repeatability

The development split and two post-development workload campaigns consume 0.404 H100-hours excluding the three anchor runs. The balanced configuration corpus consumes 0.617 H100-hours (0.200 Probe and 0.417 Verify), while winner confirmation consumes 0.0956 H100-hours. These reported campaigns total 1.117 H100-hours, again excluding the three anchor executions. Cheap evidence does not erase this offline cost; it establishes the controlled corpus needed to decide which costs a later online policy can avoid.

6 Discussion

What the positive energy result means. The frozen blind hold-out and separately predeclared no-refit confirmation show that a sparse, interpretable correction can preserve same-H100 energy ordering across the measured context-concurrency envelope. The confirmation matters as much as the error value: it shows that the fixed rule reproduces without refitting within the tested envelope, rather than fitting the first holdout post hoc.

What the latency failure teaches. The low-concurrency TTFT failure is not an inconvenient outlier to discard. It shows that scope is metric-specific and can have internal occupancy boundaries. The correct system output is therefore a composite decision: energy supported in the tested envelope, TTFT supported only at concurrency at least four, and direct measurement required elsewhere.

Why a fidelity ladder is preferable to one predictor. The configuration experiment supplies a stronger counterexample. The CPU prior preserves energy rank but misses the frozen set’s only verified Pareto point because its TTFT ordering is reversed. Retrospectively, the short measured Probe supplies enough hardware behavior to remove the false negative, while Verify removes remaining false positives. The balanced corpus estimates these stage roles; it does not test an adaptive acquisition policy.

Separation from TokenPowerAgent. TOKENPOWERSANDBOX defines scenarios, evidence acquisition, provenance, metric-specific release, and verification. It does not decide adaptively which candidate to query next. In hardware design, FIXME provides a cross-domain methodological precedent by decomposing an end-to-end LLM-aided functional-verification workflow into five executable task subsets [19]. We similarly isolate evaluator validity before studying the policy: TokenPowerAgent will treat this interface as its environment and study adaptive acquisition, bounded language-model planning, and larger topology searches. This report proposes neither an adaptive acquisition algorithm nor a calibrated risk guarantee; its contribution is typed evidence records, a predeclared metric-scope gate, a retrospective fidelity audit, and no-reselection confirmation.

7 Related Work

Inference-energy measurement. From Words to Watts and subsequent studies quantify how model, tokens, hardware, and serving

choices alter inference energy [7, 16]. ML.ENERGY automates comparable measurement across systems, while TOKENPOWERBENCH provides declarative workloads and phase-aligned device/node telemetry [5, 14]. Other studies compare inference engines or broaden measurement across many model-GPU combinations [3, 4, 15]. These systems measure energy within a declared device, node, or system boundary. TOKENPOWERSANDBOX retains real-GPU measurement as authority but adds a freeze-before-measurement evidence contract for deciding when a cheaper record may screen future runs.

At infrastructure scale, Zhao et al. combine out-of-band cooling, rack-PDU, and node telemetry into facility-, rack-, node-, and job-aligned energy views, with optional post-run reporting through Slurm [21]. This complements our synchronized, device-only serving window: their contribution is cross-scope production observability, whereas ours is the provenance and release boundary for measured and predicted serving records. Host, rack, and cooling energy remain outside our present metric.

Workload-based energy modeling. Wilkins et al. fit offline workload-based energy and runtime models from input/output tokens on heterogeneous systems and use them for energy-optimal routing [20]. From Tokens to Watt-hours develops an analytical prefill/decode estimator for H100-class inference, while LLMCO2 learns phase-aware energy predictors across models, GPUs, and parallelism [8, 18]. Our novelty is not the existence of a token-workload regression or analytical GPU estimator. We instead study a live vLLM context-concurrency envelope, freeze predictions before a blind holdout, replicate without refitting, release metrics through explicit pass/abstain decisions, and connect workload validity to a three-stage configuration-fidelity experiment.

Serving simulation and recommendation. Vidur provides event-driven LLM serving simulation, and LLM-Pilot uses characterization to recommend model and hardware deployments [1, 11]. vLLM and related serving systems optimize batching and memory management directly [2, 10]. TOKENPOWERSANDBOX is complementary: it does not introduce a new scheduler, and it refuses to interpret one-H100 measurements as communication or queueing evidence for a different topology.

Multi-fidelity allocation. Hyperband allocates increasing resource budgets to progressively fewer configurations, while BOHB combines that racing strategy with model-based search [6, 13]. Such methods rely on lower budgets being informative for the final objective. Our balanced Probe/Verify corpus instead audits that premise under device-energy and throughput objectives with latency feasibility; it neither proposes nor evaluates an adaptive budget allocation policy.

Energy-aware serving optimization. DynamoLLM designs inference clusters under performance and energy objectives; BEAM jointly tunes resource and power controls under SLOs; BOUTe uses multi-objective Bayesian optimization for heterogeneous serving [9, 12, 17]. Such methods search a configuration space given an evaluator. This report tests that evaluator and its release boundary. Adaptive acquisition policies can be compared later on the same hash-frozen evaluation summaries without allowing a policy to

relabel simulation as measurement or train on final confirmation data.

8 Limitations and Threats to Validity

The evidence covers one Qwen2.5-7B model, one H100 80GB, one vLLM release, BF16, a fixed 700-W power limit, and two bounded protocols. The workload-transfer model fixes the serving configuration and output length; the configuration study fixes the workload. The cross-product is not explored. Six development workloads are small for a four-feature residual, and the low-concurrency TTFT failure demonstrates that a simple numerical range is not a complete scope description.

DCGM supplies device energy, not host, network, cooling, facility, or embodied energy. Startup and warmup are intentionally outside the active-window metric; deployment amortization may change the preferred configuration. Fixed token counts and temperature zero control completed work but do not test semantic quality, stochastic output lengths, finite-rate arrivals, or production traces. All requests succeeded, so failure-heavy regimes are absent.

The diagnostic prediction intervals were not calibrated before the post-development campaigns; their containment cannot be reported as statistical coverage. The independent winner campaign contains five pairs, enough to obtain the reported exact sign-test p -value but not for broad effect generalization. The observed 1.39% mean paired device-energy saving is small and should not be extrapolated beyond the tested pair.

The residual-fitting path leaves its generic automatic-release flags unset, and the campaign validator does not promote them after holdout evaluation. We accordingly make no interval-coverage claim. Separately, the inherited profile contains placeholder cross-topology values, but those terms are disabled for the reported same-H100 point estimates. We make no cross-topology claim.

Finally, the configuration CPU prior is an out-of-scope analytical extrapolation after knobs change. It is included as a transparent negative baseline, not as evidence that the workload residual transfers across serving configurations. Multi-GPU communication, placement, H200/B200 adaptation, and multi-node validation remain unmeasured. The three anchor repeats are retained only as an aggregate with dispersion and repeat count; all other 151 serving executions used here have row-level records.

9 Conclusion

TOKENPOWERSANDBOX turns cheap LLM-serving predictions into a hash-tracked evidence workflow rather than treating them as hardware truth. A frozen blind single-H100 holdout and a separately predeclared no-refit confirmation obtain 6.23% and 7.35% energy MAPE with strong rank preservation, while a predeclared gate correctly abstains from inaccurate low-concurrency TTFT. Across the frozen serving configurations, the CPU prior preserves energy order but misses the set’s verified frontier; retrospective comparison shows that a short measured Probe recovers that point, while full Verify removes false positives. A separately frozen paired campaign confirms the selected configuration with a mean paired device-energy saving of 1.39% and a median paired P95-TTFT reduction of 21.43% relative to the predeclared expert baseline in the tested scope.

The practical lesson is simple: a cheap model is useful only when its evidence kind, scope, chronology, and abstention behavior travel with every prediction, and when the final decision is independently measured. This single-GPU report establishes that substrate for the next TokenPowerAgent stage without claiming the adaptive agent or multi-GPU results in advance.

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